IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2020

WAVELETS AND APPLICATIONS SOLUTIONS

SOLUTIONS

1. (a)

$$Y(z) = \frac{1}{2}\left[G(z^{1/2})H(z^{1/2}) + G(-z^{1/2})H(-z^{1/2})\right] X(z).$$

Therefore PR condition is:

$$G(z)H(z) + G(-z)H(-z) = 2$$

or

$$P(z) + P(-z) = 2$$

(b) We try $G(z) = az^{-1} + b + az$. We have

$$P(z) = (z^{-2} + z^{-1} + 1 + z + z^2)(az^{-1} + b + az).$$

The half-band condition $P(z) + P(-z) = 2$ then leads to $a + b = 0$ and $2a + b = 1$. So the final solution is

$$G(z) = z^{-1} - 1 + z.$$

(c) We want $P(z) + P(-z) = 0$. If we pick $G(z) = (1 - z^{-1})$ then we have that $P(z) = z^3 - z^{-1}$ which clearly leads to $P(z) + P(-z) = 0$.

(d) i. The easiest way to address the question is by observing that if we denote with $g_1[n]$ the filter $h_1[0], h_1[2], h_1[4]$...., then $H_1(z) = G_1(z^2)$ and similarly we can introduce $G_2(z^2) = H_2(z)$. It then follows that $Y(z) = G_1(z)G_2(z)X(z)$.

ii. The answer is that we achieve perfect reconstruction. Again there are various ways to show that. One way is by observing that the ideal low pass filters have only even numbered non-zero coefficients and that the downsampled versions $G_1(z)$ and $G_2(z)$ are both ideal all pass filters. Therefore $G_1(z)G_2(z) = 1$.

2. (a) Two-scale equations:

$$\varphi(t) = \sqrt{2}\sum_n g_0[n]\varphi(2t-n)$$

and

$$\tilde{\varphi}(t) = \sqrt{2}\sum_n g_0[n]\tilde{\varphi}(2t-n)$$

Moreover,

$$\langle \tilde{\varphi}(t), \varphi(t-n)\rangle = \delta[n]. \tag{1}$$

By replacing $\tilde{\varphi}(t)$ and $\varphi(t-n)$ above with the corresponding two-scale equations and by using the linearity of the inner product, we obtain:

$$\begin{aligned} \langle \tilde{\varphi}(t), \varphi(t-n)\rangle &= 2\textstyle\sum_k h_o[k]\sum_l g_0[l]\langle \tilde{\varphi}(2t-k), \varphi(2t-2n-l)\rangle \\ &= 2\textstyle\sum_k\sum_l h_o[k]g_0[l]\frac{1}{2}\langle \tilde{\varphi}(x-k), \varphi(x-2n-l)\rangle, \end{aligned} \tag{2}$$

where in the last equality we have replaced $2t$ with x. Because of the duality condition in Eq. (1), we have that

$$\langle \tilde{\varphi}(x-k), \varphi(x-2n-l)\rangle = \delta[k-(2n+l)].$$

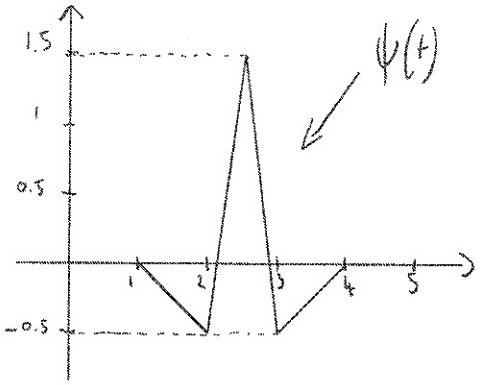
Therefore Eq. (2) becomes:

$$\begin{aligned} \langle \tilde{\varphi}(t), \varphi(t-n)\rangle &= \textstyle\sum_k\sum_l h_o[k]g_0[l]\delta[k-(2n+l)] \\ &= \textstyle\sum_l h_o[l+2n]g_0[l] \\ &= \langle g_0[l], h_0[l+2n]\rangle. \end{aligned}$$

By comparing the last term above with Eq. (1), we arrive at the desired result:

$$\langle g_0[l], h_0[l+2n]\rangle = \delta[n].$$

(b) The wavelet function



(c) The number of vanishing moments of the dual wavelet depends on the number of zeros at $z = 0$ in $G_0(z)$. Since $G_0(z)$ has two zeros then the dual wavelet has two vanishing moments.

(d) We know that we say that the restriction of $f(t)$ to $[a, b]$ is uniformly Lipschitz $\alpha \geq 0$ over $[a, b]$ if there exists $K > 0$ such that for all $\nu \in [a, b]$ there exists a polynomial $p_\nu(t)$ of degree $m = \lfloor \alpha \rfloor$ such that

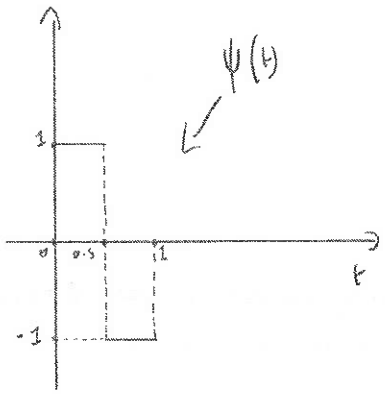
$$\forall t \in (a, b), \quad |f(t) - p_\nu(t)| \leq K|t - \nu|^\alpha. \tag{3}$$

Now, since $f(t)$ is uniformly α-Lipshitz with $\alpha = 1.6$ and since $\tilde{\psi}(t)$ has $2 = \lfloor \alpha \rfloor + 1$ vanishing moments and is of compact support. We have that the wavelet coefficients in the cone of influence of t_0 are then given by:

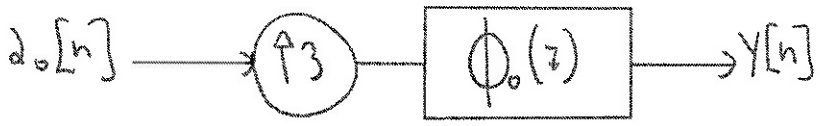
$$\begin{aligned}
\langle f, \tilde{\psi}_{m,n} \rangle &= \underbrace{\langle p_{t_0}(t), \tilde{\psi}_{m,n}(t) \rangle}_{=0} + \langle \epsilon(t), \tilde{\psi}_{m,n}(t) \rangle \\
&\overset{(a)}{\leq} K 2^{-m/2} \int_{-\infty}^{\infty} |t - t_0|^\alpha \tilde{\psi}(2^{-m}t - n) dt \\
&= K 2^{m/2} \int_{-\infty}^{\infty} |x 2^m + n 2^m - t_0|^\alpha \tilde{\psi}(x) dx \\
&\overset{(b)}{\leq} K C 2^{m(\alpha+1/2)} \underbrace{\int_{-\infty}^{\infty} (|x| + |C|)^\alpha \tilde{\psi}(x) dx}_{=A} \\
&= C_1 2^{m(\alpha+1/2)}
\end{aligned}$$

where (a) follows from (3) and (b) from the fact that we are in the cone of influence of t_0 and therefore $|n2^m - t_0| \leq C2^m$ where we have denoted with C the support of $\tilde{\psi}$.

(e) The new wavelet is indeed the Haar wavelet sketched below and has only one vanishing moment



3. (a) The system is first upsampling the sequence by a factor three and then filtering with $\Phi_0(z)$.



(b) The system (sketched below) performs an orthogonal projection onto the subspace $V =$ span $\{\phi_0(n-3k)\}_{k\in Z}$.



(c) i. We want $\langle \phi_i[n], \phi_j[n-3k] \rangle = \delta_{i-j,k}$, for $i,j = 1,2,3$. We note that, since the three filters have all support 3, the above condition is satisfied when $k \neq 0$. We therefore only need to design $\phi_1[n]$ and $\phi_2[n]$ such that

$$\langle \phi_i[n], \phi_i[n] \rangle = \delta_{i-j}, \text{ for } i,j = 0,1,2$$

Symmetry implies that $\phi_1[0] = \phi_1[2]$, so

$$\langle \phi_1[n], \phi_1[n] \rangle = 1 \Rightarrow \phi_1^2[1] + 2\phi_1^2[2] = 1$$

and

$$\langle \phi_0[n], \phi_1[n-3k] \rangle = 0 \Rightarrow \phi_1[1]/\sqrt{3} + 2\phi_1[2]/\sqrt{3} = 0.$$

This yields (solution unique up to a sign) $\phi_1[1] = 2/\sqrt{6}$ and $\phi_1[0] = \phi_1[2] = -1/\sqrt{6}$. The assumption $\phi_2[0] = -\phi_2[2]$ together with the orthogonality conditions yield $\phi_2[1] = 0$, $\phi_2[0] = 1/\sqrt{2}$ and $\phi_2[2] = -1/\sqrt{2}$.

ii. The two sequences are obtained by filtering $x[n]$ with the time-reversed versions of ϕ_1 and ϕ_2 and then subsampling by three. This yields the following expressions:

$$\alpha_1[n] = (-x[3n] + 2x[3n+1] - x[3n+2])/\sqrt{6}$$

and

$$\alpha_2[n] = (x[3n] - x[3n+2])/\sqrt{2}$$

4. (a)

$$Y_0(z) = \frac{1}{2}[H_0(z^{1/2})X(z^{1/2}) + H_0(-z^{1/2})X(-z^{1/2}]$$

and

$$Y_1(z) = \frac{1}{2}[H_1(z^{1/2})X(z^{1/2}) + H_1(-z^{1/2})X(-z^{1/2}].$$

Therefore

$$\hat{X}(z) = \frac{G_0(z)}{2}[H_0(z)X(z) + H_0(-z)X(-z)] + \frac{G_1(z)}{2}[H_1(z)X(z) + H_1(-z)X(-z)].$$

This implies that PR is achieved when

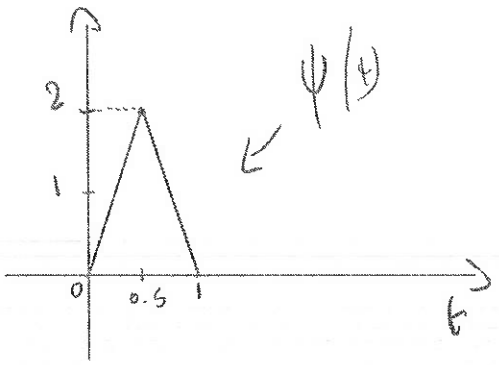
$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2$$

and

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0.$$

(b) Given that $G_0(z) = (z + 2 + z^{-1})/(2\sqrt{2})$ and that $P(z) = \frac{1}{2}(1 + z)(1 + z^{-1})$ then the only possible solution for $H_0(z)$ is $H_0(z) = \sqrt{2}$. This solution leads to a bi-orthogonal filter bank. Therefore $G_1(z) = -z^{-1}H_0(-z) = \sqrt{2}z^{-1}$ and $H_1(z) = zG_0(-z) = (-z^2 + 2z - 1)/(2\sqrt{2})$.

(c) The corresponding wavelet is a shifted and rescaled version of the linear spline as shown below:



(d) The analysis scaling function instead converges to the distribution $\sqrt{2}\delta(t)$

(e) PR condition in this case reduced to

$$H_0(z)G_0(z) + H_1(z)G_1(z) = 1.$$

This means that in this specific case we require $H_1(z)G_1(z) = 1 - H_0(z)G_0(z) = z^{-1} - 3 - z$. We then solve through spectral factorization noting that $H_1(z)G_1(z) = z^{-1}(z - a_0)(z - a_1)$ with

$$a_0 = \frac{-3 - \sqrt{5}}{2}$$

and

$$a_1 = \frac{-3+\sqrt{5}}{2}.$$

So possible solutions are:

$$H_1(z) = 1 \text{ and } G_1(z) = z^{-1}(z-a_0)(z-a_1),$$

$$H_1(z) = z^{-1}(z-a_0) \text{ and } G_1(z) = (z-a_1)$$

$$H_1(z) = z^{-1}(z-a_1) \text{ and } G_1(z) = (z-a_0)$$

or

$$H_1(z) = z^{-1}(z-a_0)(z-a_1) \text{ and } G_1(z) = 1$$

So the solution is clearly not unique.